

Machine	Temperature	Vibration	Failure
1	H	H	Yes
2	H	L	Yes
3	M	H	Yes
4	L	L	No
5	L	M	No
6	M	L	No
7	H	M	Yes
8	L	H	No

Temp = H :
Vib = M :

4 Yes 4 No Total = 8
Entropy

$$S = -\sum p_i \log_2 p_i$$

$$= -\frac{4}{8} \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8}$$

$$= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

Step 2: Information Gain

Temperature = High
3 Yes 0 No

$$E = 0$$

Temperature = Medium
1 Yes 1 No

$$E = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2}$$

$$= 1$$

Temperature = Low
3 No

$$E = 0$$

$$WS = 1 \times \frac{2}{8} = \frac{1}{4} = 0.25$$

$$IG = S - E(S, T)$$

$$= 1 - 0.25 = 0.75$$

$$0.75 > 0.061$$

∴ Root Node is Temperature

Attribute 2: Vibration
High

2 Yes 1 No

$$S = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}$$

$$= 0.918$$

Medium

1 Yes 1 No

$$S = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2}$$

$$= 1$$

Low

2 No 1 Yes

$$S = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}$$

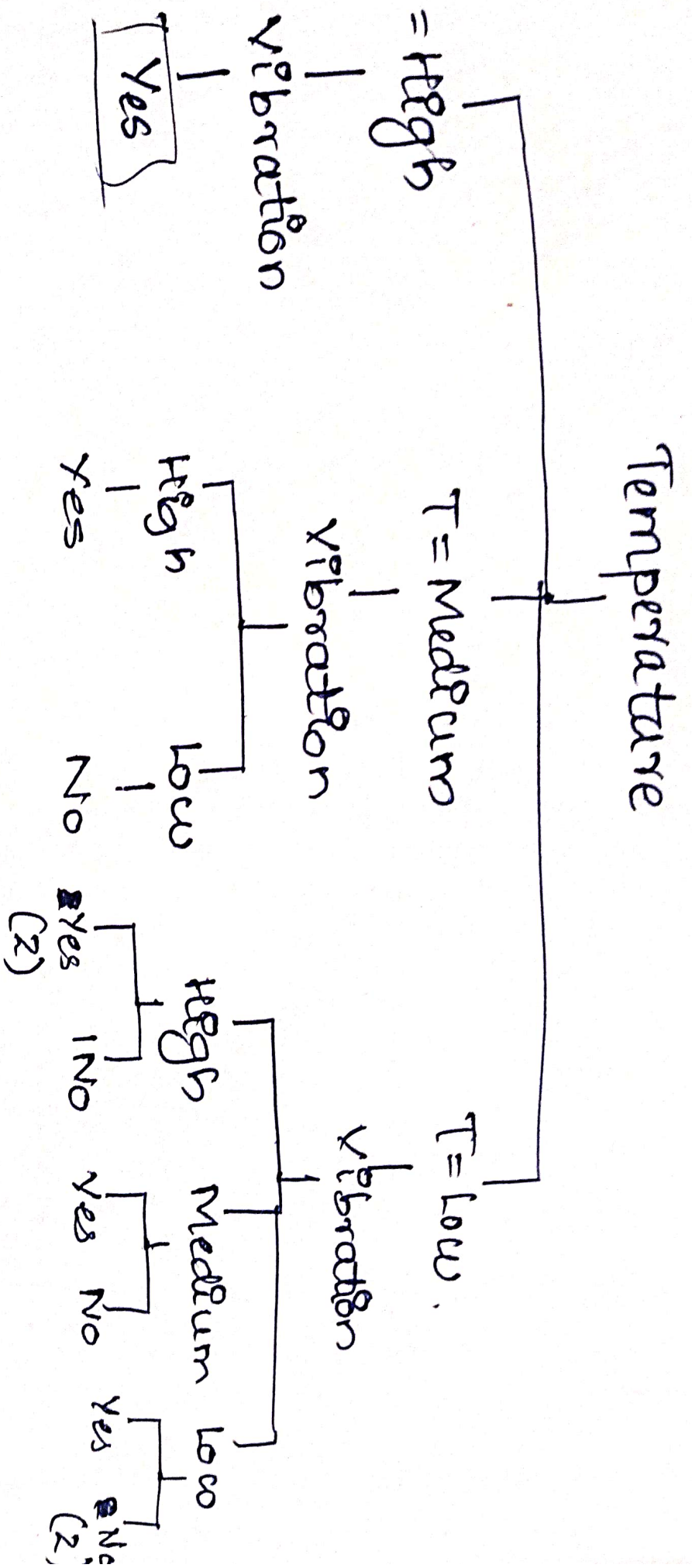
$$= 0.918$$

$$WS = 0.918 \times \frac{3}{8} + 1 \times \frac{1}{8} + 0.918 \times \frac{3}{8}$$

$$= 0.939$$

$$IG = 1 - 0.939$$

$$= 0.061$$



Temp = High
 Vibration = Medium $\rightarrow$ Yes
 Machine will fail